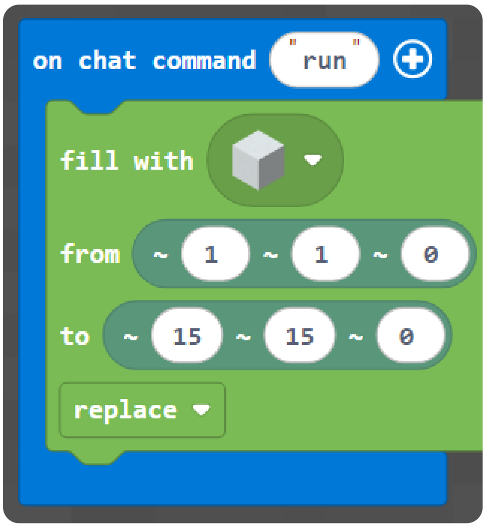
M003 Extension — English Worksheet

Topic: Pixel-Art Mobs (x, y) · **Course:** 3D Coordinates · **Level:** Extension (after Week 3) · **Time:** about 45 minutes

This is a **bonus picture challenge** between Week 3 and Week 4. You already copied one small picture using **(x, y)**. Now you copy **two bigger mobs** onto your wall — a **pink axolotl** and a **black ender dragon head**. Same idea: every block sits at a spot named by **two numbers — (x, y)**. x is how far **across**, y is how far **up**.

First, build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
        pos(1, 1, 0),
        pos(15, 15, 0),
        FillOperation.REPLACE)
player.on_chat("run", on_run)
```

> 🔴 **Big idea:** (x, y) is counted from your **home spot** — not from the corner of the world. Stand on the red block *every* time you run, or the whole picture shifts. Move your feet, move your picture.

1 · Predict

Read each set of steps. Before you imagine it happening, write what you think you will see.

```
place pink block at (4, 13)
place pink block at (5, 13)
place pink block at (6, 13)
```

Three pink blocks share y = 13. Do they make a line going across or going up? Which number stays the same?

```
place black block at (8, 7)
place black block at (8, 8)
place black block at (8, 9)
place black block at (8, 10)
```

These four share x = 8. Are they a row going across, or a column going up? Which number changes?

```
place pink block at (6, 10)
place black block at (6, 10)
```

Two blocks aim for the same **(x, y). What happens on the wall — do you see pink, black, or both? Why?**

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — This should make the **axolotl's two eyes**, one at (6, 10) and one at (11, 10). Right now both land in the **same spot**.

```
place black block at (6, 10)
place black block at (6, 10)
```

Hint: the two eyes are at **different** x values — one is further across than the other.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The dragon's **eye** should sit at (8, 10). Right now the two numbers are **swapped**, so it lands in the wrong place.

```
place purple block at (10, 8)
```

Hint: the **first** number is across (x), the **second** is up (y). Which one should be 8, which should be 10?

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — A friend stood on the **wrong block** and ran the code. The axolotl came out fine, but it was **shifted two blocks too high** — every block landed at a y two bigger than the grid says. The code is correct.

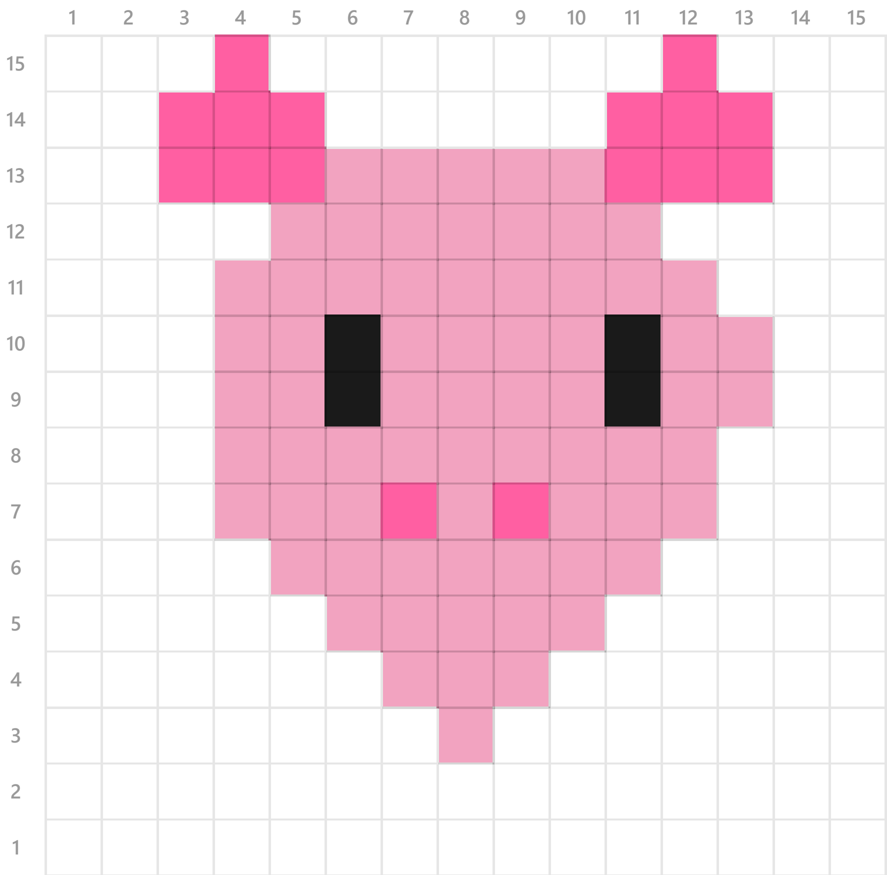
```
(the code matched the grid exactly)
```

What did your friend forget to do before pressing run? Why does standing on the home spot matter?

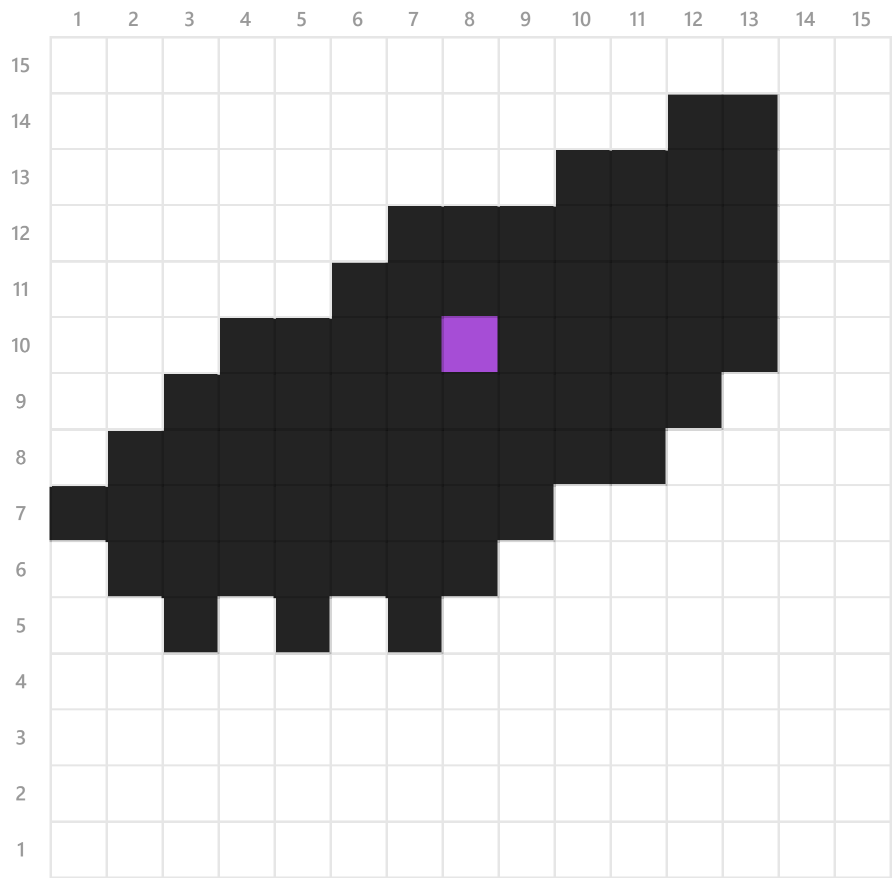
3 · Tell Me What You Built 📸

Stand on your **home spot** (the red block). Pick **one** mob below and copy it onto your wall, block by block, by reading the (x, y) of every colored square. The numbers across the **top** are **x**, the numbers down the **side** are **y**. Finished early? Build the other one too.

🦜 Axolotl (pink · #f2a3c0, frills · #ff5fa2, eyes · black)



🐉 Ender Dragon (black · #232323, eye · purple #a64dd6)



When your picture is finished, send a photo of it, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I chose to build the …

The lowest block in my picture is at (x = …, y = …).

To go across a row I changed the … number; to go up I changed the … number.

The trickiest part to read off the grid was …

One time my picture landed in the wrong place because …

If I had more time, I would …

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you show your mob on the wall. Talk through it like you are teaching someone who has never seen it. Try to use these words: **x**, **y**, **coordinate**, **across**, **up**, **home spot**.

> 1. Show your home spot (the red block) and say why you stand on it every time.
> 2. Show your finished mob from the front.
> 3. Point at one block and read its coordinate out loud: "this one is at x ..., y ...".
> 4. Show one block that you placed in the wrong spot at first, and how you fixed it.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✅

Send this worksheet + your walkthrough video to teacher on KakaoTalk.